



▲ 1844 Irish doctor Francis Rynd invents a syringe.



▲ 1849 English-born Elizabeth Blackwell graduates from medical school in the United States and becomes the first woman doctor.



▲ 1860 Pioneering French scientist Louis Pasteur (and later, German doctor Robert Koch) proves that bacteria and other microorganisms cause infectious disease.



▼ 1901 Austrian scientist Karl Landsteiner identifies blood types—later called A, B, AB, and O—enabling safe blood transfusions to take place.

1835 ————— 1850 ————— 1900



1838 German scientists Theodor Schwann and Jakob Schleiden state that all living things are made from cells.

▲ 1853–56 The work of Florence Nightingale and Mary Seacole during the Crimean War changes how wounded soldiers are nursed and leads to the establishment of modern nursing practices.

▼ 1895 X-rays are discovered by scientist Wilhelm Roentgen. The first X-ray image is of his wife's hand.



▼ 1865 To reduce wound infection by bacteria, British surgeon Joseph Lister introduces antiseptic (germ-killing) sprays into his operating theater, dramatically reducing the death rate.



▲ 1928 Scottish scientist Alexander Fleming discovers penicillin, the first antibiotic drug.